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Propositional Logic

Chapter Learning Outcomes

CLO

8

- Determine if a given sentence is a proposition or not.
- Complete and use truth tables for expressions involving the following logical connectives: negation, conjunction, disjunction, conditional, and biconditional.
- Write the converse, contrapositive and inverse for a given sentence.
- Translate an English sentence to PL and vice-versa.
- Define and determine validity, inconsistency, and satisfiability for a PL expression.
- Prove if two propositions are logically equivalent or not.

Propositional Logic

The field of mathematics known as symbolic logic, in which symbols are used to replace ordinary language.

Proposition

A **proposition**, or *statement*, is a declarative sentence that can be classified as either true or false but not both.

Propositional Logic

Proposition

A **proposition**, or *statement*, is a declarative sentence that can be classified as either true or false but not both.

Example : All of the following are **propositions**:

a) This course is discrete mathematics.
b) Today is Friday.
c) Amman is the capital of Jordan.
d) $1 + 0 = 1$
e) $0 + 0 = 2$

Propositional Logic

Proposition

A **proposition**, or *statement*, is a declarative sentence that can be classified as either true or false but not both.

Example : All of the following are **not propositions**:

a) Sit down!
b) What is the time now?
c) $x + 1 = 2$
d) $x + y = z$
e) Either 7 is an odd number or it is even

Propositional Logic

The words that are used to combine propositions are called logical connectives. The basic connectives that we will consider are given in Table 1 along with their symbols.

TABLE 1

Name	Logical Connective	Symbol
Conjunction	and	$\wedge$
Disjunction	or	$\vee$
Negation	not	$\sim$
Conditional	if . . . then	$\rightarrow$
Biconditional	if and only if	$\leftrightarrow$

Conjunction

Conjunction

A **conjunction** is a statement of the form "p and q" and is represented symbolically by

$$p \wedge q$$

The conjunction $p \wedge q$ is true if *both* p and q are true; it is false otherwise.

p	q	p ∧ q
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

Disjunction

A **disjunction** is a proposition of the form "p or q" and is represented symbolically by

$$p \vee q$$

The disjunction $p \vee q$ is false if *both* statements p and q are false; it is true in all other cases.

p	q	p Vq
T	T	T
T	F	T
F	T	T
F	F	F

Negation

Negation

A **negation** is a proposition of the form "not p" and is represented symbolically by

$$\sim p$$

The proposition $\sim p$ is true if p is false and vice versa.

p	¬p
T	F
F	T

Conditional Statement

Conditional Statement

A **conditional statement** is a proposition of the form "if p, then q" and is represented symbolically by

$$p \rightarrow q$$

The connective "if . . . then" is called the **conditional connective**; the proposition p is called the **hypothesis**; and the proposition q is called the **conclusion**. A conditional statement is false if the hypothesis is true and the conclusion is false; it is true in all other cases.

p	q	p →q
T	T	T
T	F	F
F	T	T
F	F	T

Different Forms for Implications

"if H then C" "$H \rightarrow C$" is equivalent to :

1- if H, C

2- H implies C

3- H only if C

4- C if H

5- C when H

6- C whenever H

Biconditional Statement

Biconditional Propositions

Statements of the form "p if and only if q" are called **biconditional propositions** and are represented symbolically by

$$p \longleftrightarrow q$$

The connective "if and only if" is called the *biconditional connective*. The biconditional $p \longleftrightarrow q$ is true whenever p and q are *both true* or *both false*.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Same as **Exclusive-NOR**

$\overline{A \oplus B}$

Different Forms for Biconditional

"**p if and only if q**" "**P ↔ q**" is equivalent to :

1- *p* **iff** *q*

2- *p* **is necessary and sufficient for** *q*

3- if p then q , and conversely

Translating English Sentences to PL

Amman is the capital of Jordan and it is the largest city.

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Translating English Sentences to PL

Ali likes fishing and reading.

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Translating English Sentences into PL

Jones is a fool **or** a liar

Jones is **neither** a fool **nor** a liar

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Translating English Sentences into PL

Ali can not play football today.

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Translating English Sentences into PL

If it rains, then the baseball game will be postponed.

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Translating English Sentences to PL

Ali is going to the university if and only if Ahmad is going to the university.

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Translating English Sentences to PL

Let p and q be the propositions:

- p : The stock market goes up.
- q : Interest rates decrease.

Represent the following statements using PL:

1- If interest rates decrease, then the stock market goes up.

2- If interest rates do not decrease, then the stock market does not go up.

Translating English Sentences to PL

Let p and q be the propositions:

- **p : The stock market goes up.**
- **q : Interest rates decrease.**

Represent the following statements using PL:

3- The stock market will go up if and only if interest rates decrease.

Translating English Sentences to PL

If a number is divisible by 10, then it is divisible by 2

$p \rightarrow q$	
where:	
p	If a number is divisible by 10
q	It is divisible by 2

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Translating English Sentences into PL

It is raining, when Ali is come.

p when q ≡ p if q ≡ if q then p

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Order of Precedence of Logical Operators

$$(\,),\quad \neg,\quad \wedge,\quad \vee,\quad \rightarrow,\quad \leftrightarrow,\quad \equiv$$

$$\underbrace{\underbrace{p \vee q}_{2} \rightarrow \underbrace{\neg r}_{1}}_{3}$$

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Truth Tables For Compound Propositions

EXAMPLE

Construct a truth table for $p \vee q \rightarrow \neg r$.

1- How many atomic sentences are there in the expression?

1- How many compound sentences are there in the expression?

Truth Tables For Compound Propositions

Construct a truth table for $p \vee q \rightarrow \neg r$

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Truth Tables For Compound Propositions

EXAMPLE

Construct a truth table for $\neg(p \rightarrow q) \wedge p$.

1- How many atomic sentences are there in the expression?

1- How many compound sentences are there in the expression?

Truth Tables For Compound Propositions

Construct a truth table for $\neg(p \rightarrow q) \wedge p$

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Quiz

Construct a truth table for the following sentences. In each question determine the number of atomic and compound sentences.

1- $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$

2- $[(p \rightarrow q) \vee (q \rightarrow r)] \rightarrow (p \rightarrow r)$

Tautologies, Contradictions, and Contingencies

- A **tautology (valid)** is a proposition which is **always true**.
  Example: $p \vee \neg p$.

- A **contradiction** is a proposition which is **always false**.
  Example: $p \wedge \neg p$.

- A **contingency** is a proposition which **is neither a tautology nor a contradiction,** such as p.

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Tautologies, Contradictions, and Contingencies

EXAMPLE

Prove that the following proposition is tautology .

$$p \vee \neg p$$

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Tautologies, Contradictions, and Contingencies

EXAMPLE

Prove that the following proposition is contradiction.

$$p \wedge \neg p$$

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EXAMPLE

Prove that the following proposition is contingency.

$$\neg p \vee q$$

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Tautologies, Contradictions, and Contingencies

EXAMPLE

Check if the following proposition is tautology, contingency or contradictory.

$$(p \rightarrow r) \vee (q \rightarrow r)$$

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Tautologies, Contradictions, and Contingencies

EXAMPLE

Prove that the following statement is **tautology**.

$$P \wedge \neg P \rightarrow Q$$

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Quiz

EXAMPLE

Prove that the following statement is **contradiction**.

$$\neg(\neg(P \vee Q) \leftrightarrow \neg P \wedge \neg Q)$$

Converse, Contrapositive and Inverse

We refer to statements that are variations of the conditional H → C as logical variants. We define three logical variants of the conditional: H → C as follows:

• Converse	if C, then H	C → H
• Contrapositive	if not C, then not H	¬ C → ¬ H
• Inverse	if not H, then not C	¬ H → ¬ C

Converse, Contrapositive and Inverse

$$H \rightarrow C$$

Inverse: $\neg H \rightarrow \neg C$

Converse: $C \rightarrow H$

Contrapositive: $\neg C \rightarrow \neg H$

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Converse, Contrapositive and Inverse

EXAMPLE

Write the converse, the contrapositive, and the inverse for the following conditional statement:

$$p \rightarrow \neg q$$

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Converse, Contrapositive and Inverse

EXAMPLE

Write the converse, the contrapositive, and the inverse for the following conditional statement:

If n^2 is an odd number, then n is an odd number

Converse, Contrapositive and Inverse

EXAMPLE

Write the converse, the contrapositive, and the inverse for the following conditional statement:

If it rains, then they cancel the school

Logically Equivalent

Two compound propositions p and q are **equivalent** if and only if **the columns in a truth table giving their truth values agree**.

EXAMPLE

Prove that the following propositions are **equivalent**.

$$\neg p \vee q \equiv p \rightarrow q$$

P	Q	$\neg P$	$\neg p \vee q$	$P \rightarrow q$
T	T			
T	F			
F	T			
F	F			

Logically Equivalent

Prove that the following propositions are **equivalent**.

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

p	q	R	$q \vee r$	$p \wedge (q \vee r)$	$p \wedge q$	$p \wedge r$	$(p \wedge q) \vee (p \wedge r)$
T	T	T					
T	T	F					
T	F	T					
T	F	F					
F	T	T					
F	T	F					
F	F	T					
F	F	F					

Logical Equivalences

1-	Identity laws	$p \wedge T \equiv p$ $p \vee F \equiv p$
2-	Domination laws	$p \wedge F \equiv F$ $p \vee T \equiv T$
3-	Idempotent laws	$p \wedge p \equiv p$ $p \vee p \equiv p$
4-	Commutative laws	$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$
5-	Associative laws	$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
6-	Distributive laws	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
7-	De Morgan's laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Logical Equivalences

8-	Absorption laws	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$
9-	Negation laws	$p \wedge \neg p \equiv F$ $p \vee \neg p \equiv T$
10-	Implication Elimination	$p \rightarrow q \equiv \neg p \vee q$
11-	Contraposition law	$p \rightarrow q \equiv \neg q \rightarrow \neg p$
12-	Biconditional Elimination	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

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Check - Chapter Learning Outcomes

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End of Chapter

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